

Standard Operating Procedure – using MEG head casts

PURPOSE:

In order to ensure the safety and comfort of participants scanned while wearing head casts, the outlined procedures are to be followed by all researchers for all MEG scans involving head casts. The aim is for negligible risk and minimal burden to subjects by proper training and instruction.

SAFETY:

The head cast is designed to fit the subject's head internally and the MEG dewar externally. This means that the participant's head will be reasonably fixed inside the dewar and any unexpected movement of the participant has the potential to cause injury. To summarize this document:

NEVER MOVE THE MEG CHAIR OR DEWAR WITH THE PARTICIPANT'S HEAD IN THE MEG WHILE WEARING A HEAD CAST

RESPONSIBILITIES:

- **Researchers** must be MEG and head cast certified user before applying the MEG head cast procedure on a subject.
- **The DCCN Principle Investigator** of the study ensures that the researcher is a MEG and head cast certified user before starting subject inclusion.
- **The MEG lab manager** is responsible for head cast training, certification and keeping the content of the documentation up-to-date.

KEY WARNINGS:

- The dewar and chair (i.e. all moving parts) should remain locked in position whenever the subject is wearing the head-cast. This will minimize risk of neck injury.
- The chair should be positioned directly against the gantry.
- The head casts should only be used with healthy volunteer subjects. The researcher must have done two sessions without problems during prime time before the head casts may be used outside of prime time hours. Outside of prime time hours a buddy must be present.
- The head casts should only be used while subject is seated – The use of the head cast in supine position is only allowed if there is an extra support system to keep the bed in place.

GENERAL PROCEDURES:

1. Only certified MEG users who have completed the head cast specific safety training are certified to handle subjects during MEG scans involving head casts.
2. Subjects must be informed about the use of MEG head casts prior to their visit.
3. Prior to inclusion and head cast construction, subjects must be informed about the use of the head cast and, on their first visit, introduced to the procedures using the MEG dewar and head cast model.
4. Before the participant is placed in the MEG, the MEG dewar dummy model is used to practice going in and out of the scanner. If the participant is not able to easily come out of the MEG dewar model the experiment stops there and the scanning will not take place.
5. As usual, all study participants will be explicitly informed that they can withdraw from participation at any time without further explanation.

EXTRA OPERATION STEPS FOR A MEG HEAD CAST EXPERIMENT:

Subject Preparation (Initial Session):

- MRI Scan
- Subject is informed about the MEG and head cast procedure

Subject Preparation **in control room**:

- The researcher checks if there is a hole in the top of the head cast. The hole in the head cast is there to prevent a vacuum effect.
- Talcum powder will be placed on the exterior of the head cast to minimize friction with the interior of the dewar.
- Properly instruct subject how to put on the head cast outside of the scanner.
- Allow subject to wear the head cast outside of the scanner for proper fit.
 - If the head cast is too tight the head cast can be adjusted under supervision of the TG. There is a special toolbox for this in the office of Uriel.
- Subject must practice exiting the dewar dummy model by pressing in and pulling down the head cast earflaps.
 - **If the subject is not able to exit the head cast by themselves the experiment stops there.**
- Explain how the subject has to enter and exit the dewar while wearing a head cast.
- Provide opportunity for questions & any concerns before continuing.
- Subject maintains right to withdraw participation at any time without further explanation

Subject Preparation in **MSR**:

1. Without head cast on:

- Chair is at the lowest position
- Lock all seat operation valves
- Have the participant seated
- Unlock the seat raise valve
- Raise the seat until the participants head touches the dewar
- Lock the seat raise valve
- The participant “crawls” out of the seat

2. The participant puts on his head cast – when the head cast is on all moving parts must be locked

3. With the head cast on:

- The participant slides back in the seat and puts his head back in the dewar
- If the seat height needs adjustment, see procedure 4
- Practice sliding in and out the dewar
 - **If the subject is not able to exit the dewar by themselves the experiment stops there**
- Slide out of the dewar
- Place head localization coils
- Put head back in the dewar
- The shielded room will be sealed
- Researcher must always see and hear the subject
- The experiment can now begin

4. If the seat height needs adjustment:

- the participant “crawls” out of the seat,
- Participant gets back in the seat **without head cast**
- Unlock the seat raise or seat lower valve
- Adjust the height
- Lock the seat raise or lower valve
- Participant “crawls” out of the seat
- Participant puts on the head cast
- Participant gets back in the seat and puts head in the dewar

Appendix: Head Cast Safety Training Checklist

In addition to the standard MEG certified user checklist, the following are important:

- ☐ Researcher has completed standard MEG certified user checklist
- ☐ Researcher has read/understood the MEG Head Casts Standard Operating Procedure (SOP)
- ☐ Researcher has demonstrated they can competently follow the head cast SOP & understands the key dangers
- ☐ Researcher is familiar with all subject monitoring equipment (i.e. video & audio) & signs the log-book with the understanding that this signifies all monitoring was maintained throughout experiment
- ☐ Researcher is familiar with emergency procedures (emergency contact numbers)
- ☐ Researcher is aware of how to screen subjects for head casts (when to use, if unsure DO NOT continue)
- ☐ Researcher has demonstrated ability to fit a participant's head cast outside of the scanner using the dewar template & can competently instruct a subject in all experimental procedures
- ☐ Researcher understands when to stop the experiment

Change log

Version 1.2

- Head casts can be used in supine position if extra support system for the bed is used.